

Tarea - Coeficiente de expansión térmica

$$\Delta D = D \int_{T_{room}}^{T_{liq}} \alpha(T) dT \quad \text{con} \quad \alpha(T) = a_0 + a_1 T + a_2 T^2$$

$$\bullet \quad a_0 = 1.0827 \times 10^{-5} / ^\circ\text{C} \quad a_1 = 1.11503 \times 10^{-8} / ^\circ\text{C}^2$$

$$a_2 = -2.21004 \times 10^{-11} / ^\circ\text{C}^3$$

$$\bullet \quad \Delta D = D \int_{T_r}^{T_f} \alpha(T) dT = D \int_{T_r}^{T_f} (a_0 + a_1 T + a_2 T^2) dT$$

$$\Delta D = D \left[a_0 T + \frac{a_1}{2} T^2 + \frac{a_2}{3} T^3 \right]_{T_r}^{T_f}$$

a) Tomando $T_r = 26^\circ\text{C}$ y $T_f = -77^\circ\text{C}$ (Hielo seco):

$$\Delta D = -1.0824 \times 10^{-3} D \quad \text{ó} \quad -0.0010824 D$$

b) Tomando $T_r = 26^\circ\text{C}$ y $T_f = -196^\circ\text{C}$ (Nitrógeno Líquido):

$$\Delta D = -2.1376 \times 10^{-3} D \quad \text{ó} \quad -0.0021376 D$$